

1. Network Details

Architectures Used

- MobileNetV2**
A lightweight convolutional neural network optimized for edge deployment. It uses inverted residuals and linear bottlenecks, offering parameter efficiency.
Approximate Parameters: ~2.3M (for the multitask configuration).
- EfficientNetB0**
A compound-scaled CNN architecture that balances depth, width, and resolution. It provides stronger feature extraction compared to MobileNetV2.
Approximate Parameters: ~5.3M (for the multitask configuration).

Training Settings

- Dataset size: 3,999 balanced samples (8 classes × ~500 each).
- Input size: 224 × 224 × 3 (RGB).
- Batch size: 32
- Epochs: Maximum 100 with EarlyStopping.
- Learning rates:
 - MobileNetV2: 3e-4
 - EfficientNetB0: 1e-4 (fine-tuned last 100 layers).

Rationale for choosing baselines:

MobileNetV2 was selected as a lightweight baseline to provide fast training and lower compute cost. EfficientNetB0 was included as a stronger baseline, offering higher performance at greater computational cost.

2. Comparison of Baselines

Both models were trained with the same multitask setup:

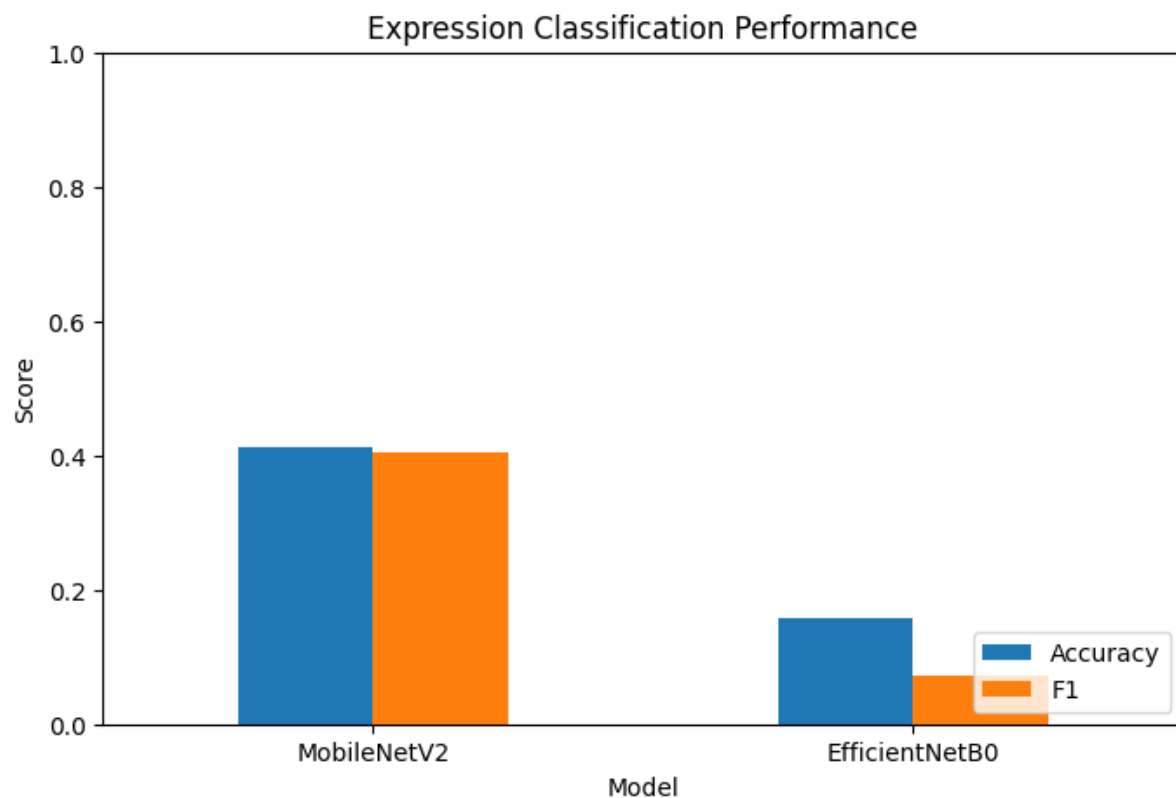
- Expression classification (8 classes)
- Valence regression
- Arousal regression

Final Performance Results:

Model	Accuracy	F1-Score	Valence RMSE	Arousal RMSE
MobileNetV2	0.4133	0.4038	0.4630	0.3989
EfficientNetB0	0.1583	0.0715	0.4653	0.4126

Observations:

- MobileNetV2 significantly outperformed EfficientNetB0 in classification tasks (accuracy and F1).
- For regression tasks (valence and arousal), both models achieved similar RMSE values, with MobileNetV2 slightly better overall.
- Despite being a smaller network, MobileNetV2 generalized better on this dataset compared to EfficientNetB0.



3. Transfer Learning

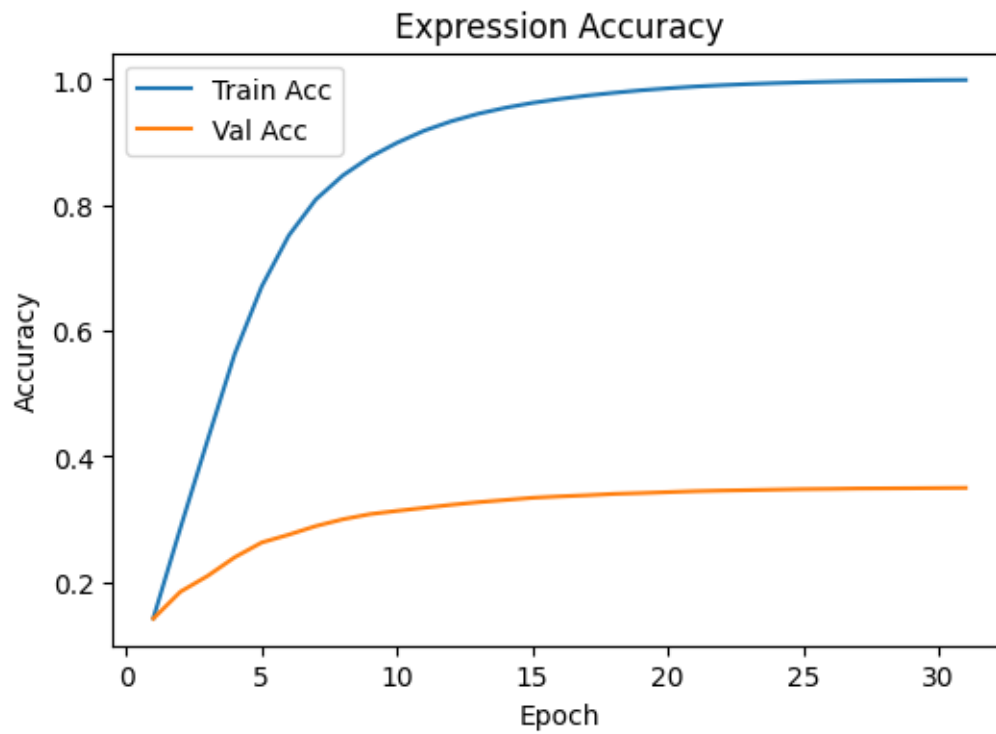
Both models used pretrained ImageNet weights. The classification and regression heads were initialized randomly. For EfficientNetB0, the last 100 layers were unfrozen to enable deeper fine-tuning, reducing class collapse and improving convergence. Transfer learning significantly improved generalization given the relatively small dataset size.

4. Training Graphs

MobileNetV2

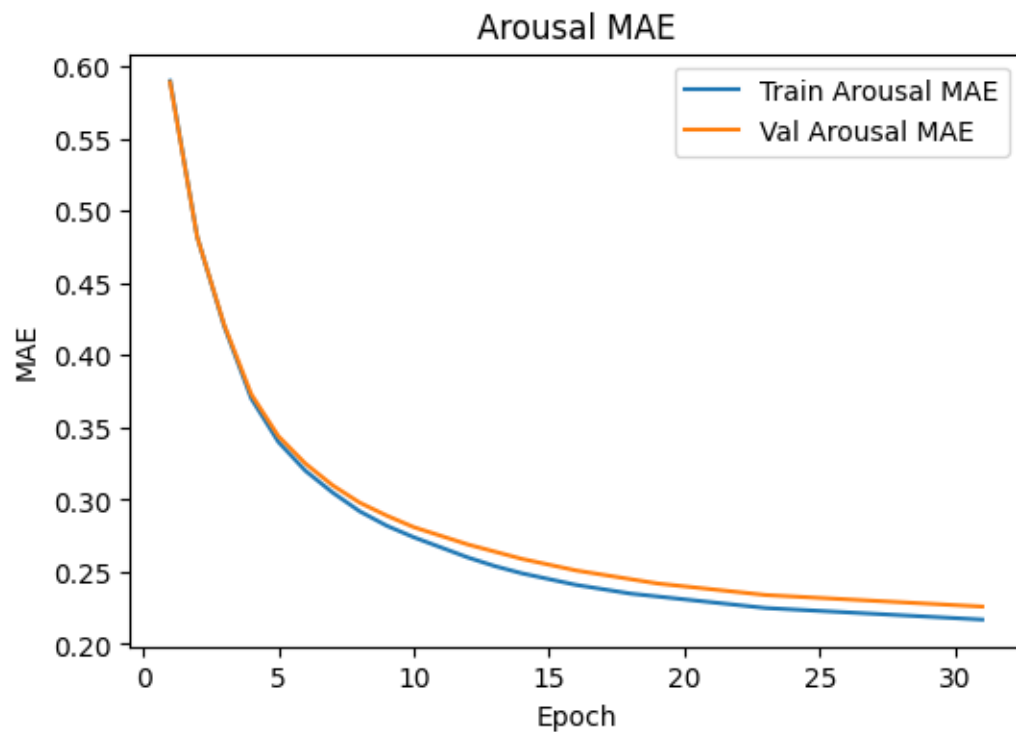
- **Expression Accuracy**

Validation accuracy steadily improved from ~12% to ~41%.



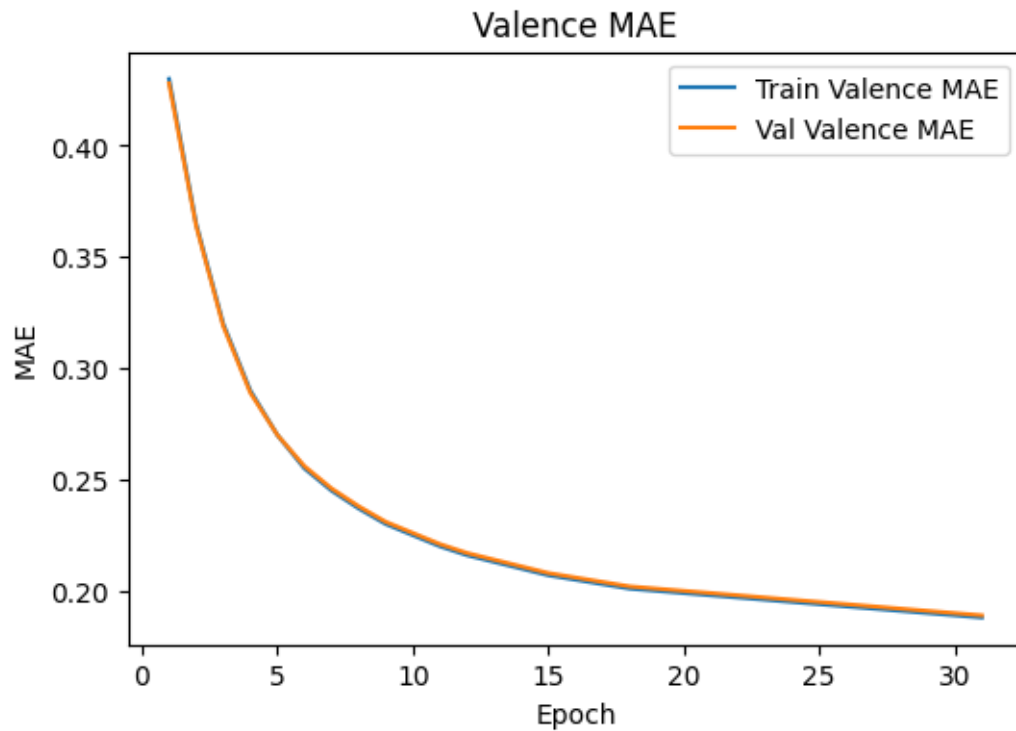
- **Arousal MAE**

Validation MAE decreased to ~0.40 by epoch 30, showing effective regression learning.



- **Valence MAE**

Validation MAE stabilized around ~0.46, showing reasonable convergence.



EfficientNetB0

- Training vs Validation Loss**

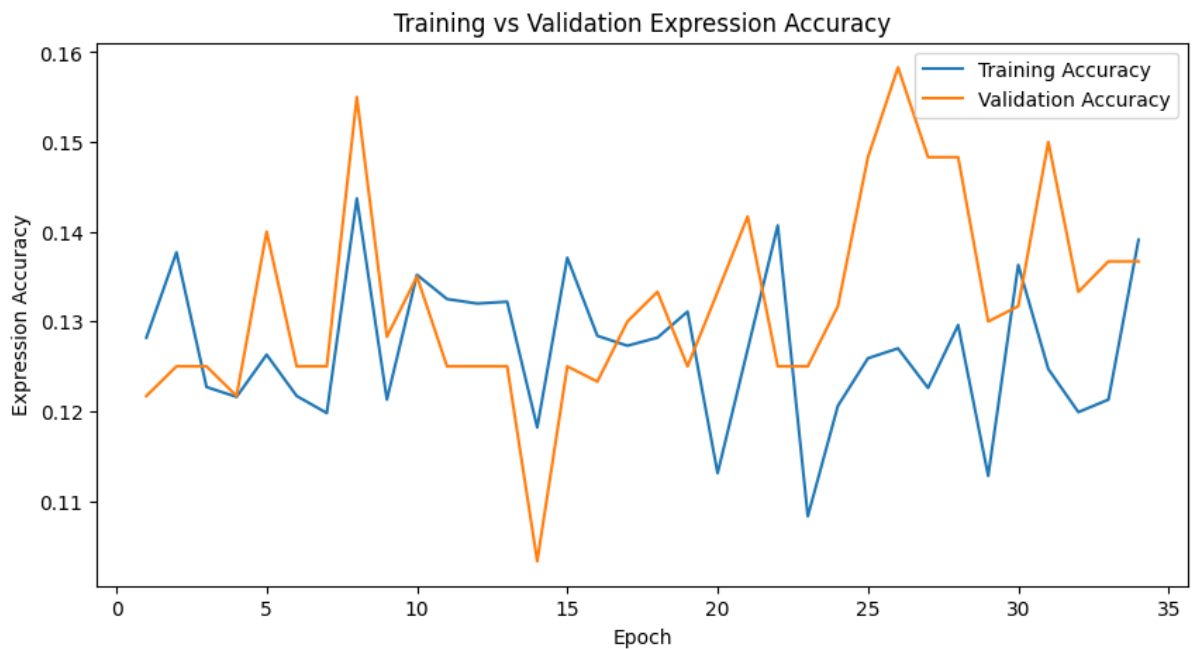
Loss curves decreased and stabilized by epoch 30, though validation performance lagged behind training.



- Training vs Validation Expression Accuracy**

Validation accuracy remained low (~15%) despite training accuracy improving,

indicating overfitting.



5. Performance Metrics and Continuous Evaluation

For classification tasks (expressions):

- **Accuracy:** Measures overall correctness.
- **F1-Score:** Balances precision and recall, handling potential class imbalance.

For regression tasks (valence, arousal):

- **RMSE (Root Mean Square Error):** Penalizes large deviations.
- **Correlation (CORR):** Captures linear association between predictions and ground truth.
- **SAGR (Sign Agreement Metric):** Ensures correctness of polarity (positive/negative), which is critical for affective computing.
- **CCC (Concordance Correlation Coefficient):** Combines correlation and bias, offering the most balanced evaluation.

```
• Evaluating MobileNetV2...
• Valence:
• RMSE: 0.4629986311056033
• CORR: 0.53377676
• SAGR: 0.5333583333333334
• CCC : 0.5111346
• Arousal:
• RMSE: 0.3989484166572037
• CORR: 0.38581175
• SAGR: 0.6745944444444445
• CCC : 0.37962243
• =====
• Evaluating EfficientNetB0...
```

- Valence:
 - RMSE: 0.4652663481749285
 - CORR: -0.05067377
 - SAGR: 0.6933333333333334
 - CCC : -0.0019543641
- Arousal:
 - RMSE: 0.41263250621357656
 - CORR: -0.033303082
 - SAGR: 0.775
 - CCC : -0.0014771649

Most suitable metric in the wild:

CCC is the most reliable as it accounts for both correlation and bias. SAGR is also crucial since polarity correctness is more important than magnitude alone.

6. Single Image Evaluation

Testing on images/100.jpg:

- **MobileNetV2**
 - Predicted Expression: **Surprise**
 - Valence: **-0.108**
 - Arousal: **0.496**
- **EfficientNetB0**
 - Predicted Expression: **Anger**
 - Valence: **-0.159**
 - Arousal: **0.176**

Conclusion

MobileNetV2 provided a fast and efficient baseline and performed better than EfficientNetB0 on both classification and regression tasks for this dataset. Transfer learning and fine-tuning were essential to achieving convergence. Continuous metrics such as CCC and SAGR are more reliable than raw RMSE for real-world affective systems.